

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 10 of 10

Applications of the SFEE and Unsteady Flow Processes

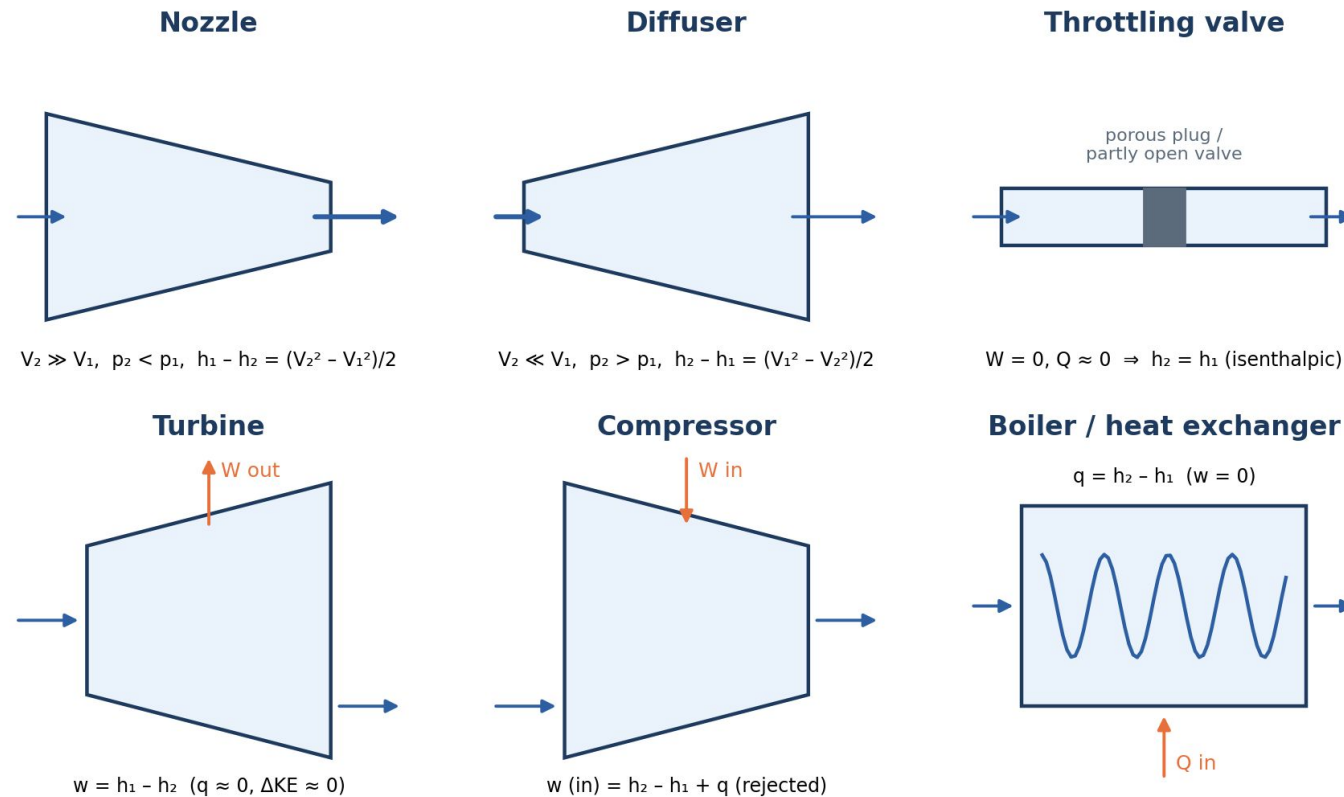
Nozzle and diffuser · turbine and compressor · boiler, condenser and heat exchanger · throttling · charging and discharging of vessels · Unit I formula sheet

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Steady-flow devices at a glance



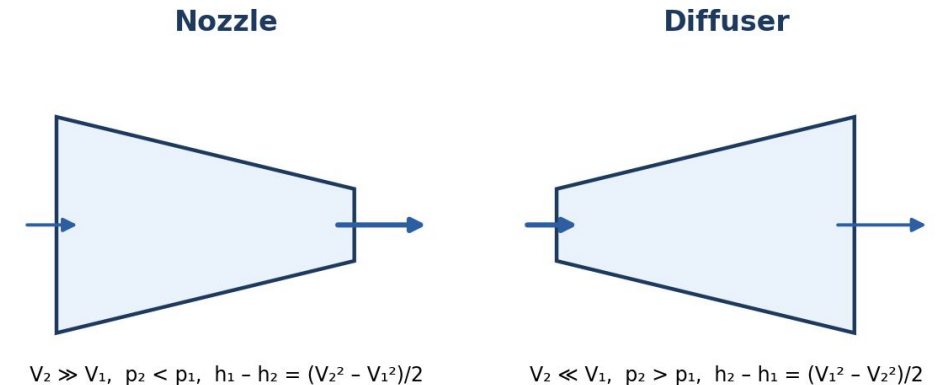
Each device is analysed by identifying which SFEE terms are negligible; ΔPE is neglected in all cases

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Nozzle and diffuser

Nozzle and diffuser

- **Nozzle** – decreasing section: velocity rises, pressure and enthalpy fall
- **Diffuser** – increasing section: fluid decelerates, pressure rises
- Short passages, no moving parts: $w = 0$
- Very little time inside: $q \approx 0$



SFEE for a nozzle or diffuser

$$h_1 + V_1^2/2 = h_2 + V_2^2/2$$

Nozzle exit velocity: $V_2 = \sqrt{2(h_1 - h_2) + V_1^2}$ (h in J/kg)

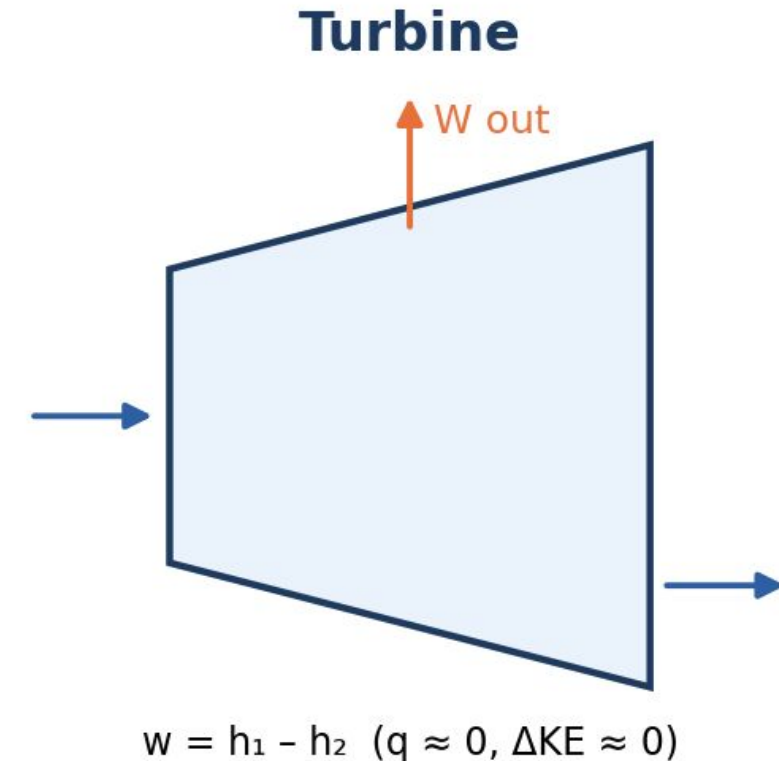
- Nozzle: enthalpy drop $\rightarrow$ kinetic energy; with negligible inlet velocity $V_2 = \sqrt{2(h_1 - h_2)}$
- Diffuser: $h_2 - h_1 = (V_1^2 - V_2^2)/2$; for a gas $T_2 = T_1 + (V_1^2 - V_2^2)/(2c_p)$
- Exit area from continuity: $A_2 = \dot{m}v_2/V_2$

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Turbine and compressor

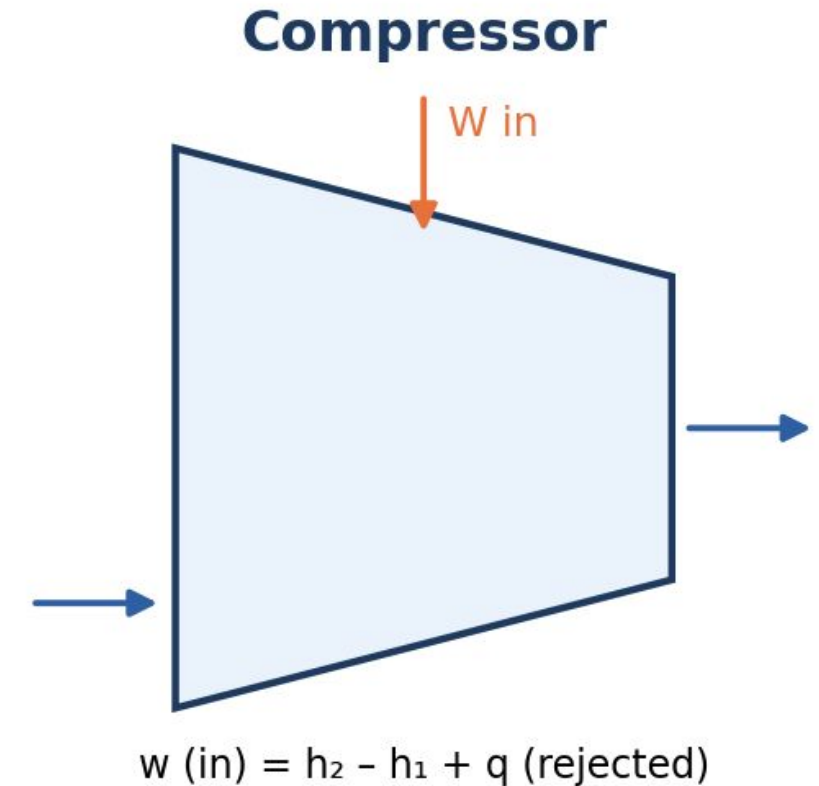
Turbine

- Produces **shaft work** from the expansion of a fluid
- Insulated casing: $q \approx 0$; inlet and exit velocities small or nearly equal: $\Delta KE \approx 0$
- $w = h_1 - h_2$; power $\dot{W} = \dot{m} (h_1 - h_2)$
- With appreciable velocities: $w = (h_1 - h_2) + (V_1^2 - V_2^2)/2$
- With casing heat loss: $w = (h_1 - h_2) - q_{\text{loss}}$



Compressor and pump

- Receives shaft work and raises the pressure of the fluid; w is **negative**
- Rotary compressor, $q \approx 0$, $\Delta KE \approx 0$: **work input** = $h_2 - h_1 = c_p (T_2 - T_1)$ for a gas
- Water-jacketed reciprocating compressor: work input = $(h_2 - h_1) + q_{\text{rejected}}$
- where $h_2 - h_1 = (u_2 - u_1) + (p_2 v_2 - p_1 v_1)$



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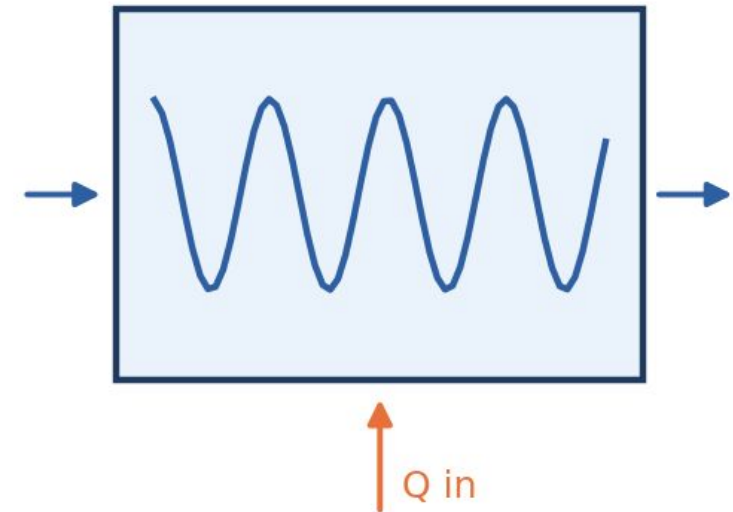
Heat-transfer devices and throttling

Boiler, condenser and heat exchanger

- No shaft work; velocities and elevation changes small
- $q = h_2 - h_1$: heat transferred per kg equals the enthalpy change of the stream
- Two-stream heat exchanger with no external loss: enthalpy lost by the hot stream = enthalpy gained by the cold stream
- $\dot{m}_h (h_{h1} - h_{h2}) = \dot{m}_c (h_{c2} - h_{c1})$

Boiler / heat exchanger

$$q = h_2 - h_1 \quad (w = 0)$$



Throttling device

- Flow through a restriction – partly open valve, orifice or porous plug – with a large pressure drop
- No work, negligible heat transfer, small and nearly equal velocities
- SFEE gives $h_1 = h_2$: throttling is **isenthalpic**
- Highly irreversible – a pressure drop with no useful effect
- Used in expansion valves, the throttling calorimeter (Unit III) and the Joule–Thomson experiment (Unit IV)

Throttling valve



$$W = 0, Q \approx 0 \Rightarrow h_2 = h_1 \text{ (isenthalpic)}$$

Reduced forms of the SFEE

Device	Assumptions	SFEE reduces to
Nozzle / diffuser	$w = 0, q \approx 0$	$h_1 + V_1^2/2 = h_2 + V_2^2/2$
Turbine	$q \approx 0, \Delta KE \approx 0$	$w = h_1 - h_2$
Compressor / pump	$q \approx 0, \Delta KE \approx 0$	$w_{in} = h_2 - h_1 (+ q_{rejected} \text{ if cooled})$
Boiler / condenser / heat exchanger	$w = 0, \Delta KE \approx 0$	$q = h_2 - h_1$
Throttling valve	$w = 0, q \approx 0, \Delta KE \approx 0$	$h_1 = h_2$

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Unsteady flow processes

Charging and discharging of vessels

The general control-volume energy balance

$$Q - W = (m_2 u_2 - m_1 u_1)_{cv} + \sum m_e h_e - \sum m_i h_i$$

- Filling a tank from a line or emptying it through a valve: mass and energy inside the CV change with time – the flow is **unsteady** (transient)
- Subscripts 1, 2: initial and final contents of the tank; i, e: entering and leaving streams, each carrying its enthalpy
- Kinetic and potential energy neglected

Charging an evacuated insulated tank

- Rigid, insulated, evacuated tank ($m_1 = 0$) filled from a large line at h_i
- $Q = 0$, $W = 0$, no exit stream: $m_2 u_2 = m_i h_i$ and $m_2 = m_i$
- Therefore $u_2 = h_i$ – tank internal energy = line enthalpy
- The flow work of the line fluid becomes internal energy – the tank fluid ends **hotter** than the line

KEY POINT Ideal gas: $c_v T_2 = c_p T_i \Rightarrow T_2 = \gamma T_i$. Air from a line at 300 K reaches 420 K (147 °C).

Discharging a tank

- Tank emptied slowly through a valve with no heat transfer
- The fluid remaining in the tank undergoes a **reversible adiabatic expansion** – the leaving fluid takes its enthalpy with it
- Pressure and temperature fall according to $pv^\gamma = C$
- Applications: air receivers, gas-cylinder filling, pressure-vessel blow-down

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Unit I at a glance

Formula sheet

Unit I formula sheet (1) – basics and work

- Closed – fixed mass; open – control volume; isolated – no mass, no energy
- Properties: point functions, $\oint = 0$; heat and work: path functions
- Quasi-static: equilibrium states; reversible: quasi-static + no dissipation
- Zeroth law $\rightarrow$ temperature; $T \text{ (K)} = t \text{ (}^\circ\text{C)} + 273.15$
- $W = \int p \, dV$; isobaric $p (V_2 - V_1)$; isochoric 0
- Isothermal $p_1 V_1 \ln (V_2/V_1)$; polytropic $(p_1 V_1 - p_2 V_2)/(n - 1)$

Unit I formula sheet (2) – first law and properties

- Cycle: $\oint \delta Q = \oint \delta W$; process: $Q - W = \Delta U$ (Q in +, W out +)
- Energy is a property; PMM1 is impossible
- $h = u + pv$; $c_v = (\partial u / \partial T)_v$, $c_p = (\partial h / \partial T)_p$
- Ideal gas: $du = c_v dT$, $dh = c_p dT$;
 $c_p - c_v = R$
- Isochoric $Q = \Delta U$; isobaric $Q = \Delta H$; isothermal $Q = W$
- Polytropic $Q = W (\gamma - n) / (\gamma - 1)$;
adiabatic $Q = 0$

Unit I formula sheet (3) – flow processes

- SFEE: $h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w$
- Continuity $\dot{m} = AV/v$; $V^2/2000$ and $gz/1000$ in kJ/kg
- Nozzle $\Delta(V^2/2) = -\Delta h$; turbine $w = h_1 - h_2$
- Compressor $w_{in} = h_2 - h_1 + q_{rej}$
- Boiler / heat exchanger $q = \Delta h$; throttling $h_1 = h_2$
- Charging an evacuated tank: $u_2 = h_{line}$, $T_2 = \gamma T_{line}$

Unit I complete

- Numerical problems: Worked Examples 1.6 – 1.11 and Practice Problems P1 – P8 in the lecture notes
- Review questions: 29 two-mark and 13 thirteen-mark questions in the notes
- Next unit: the Second Law of Thermodynamics – Unit II